

Salinity and the discharge of salts from catchments in Australia

A.J. Peck^{a,*}, Tom Hatton^{b,1}

^a*A.J. Peck and Associates, P.O. Box 1213, Subiaco, WA 6904, Australia*

^b*CSIRO Land and Water, Private Bag No 5, Wembley, WA 6913, Australia*

Abstract

Discharge of soluble salts from catchments following clearing of native vegetation for dryland agriculture is a serious environmental and economic problem affecting soil and water resources in Australia. The fundamental challenges are: To identify areas of soil at risk of becoming saline. To relate soil and water salinity risk to options for management of land in the area that contributes to the risk, and thereby contribute to the evaluation of management options.

These challenges are faced in an environment where native vegetation has roots that extend to depths of order 10 m in soil profiles that are formed by in situ weathering of granitic rocks and dolerite dykes to depths of about 20 m. The profiles typically contain 1–100 kg m⁻² of salt (primarily sodium chloride) in solution in the pore water. The distribution of soluble salts, and the movement of water within most of the unsaturated zone of these soils results from a combination of matrix flow and flow through remnant root channels and larger-scale structures with geologic origins.

Recognized options for management of salinity risk, or to reduce existing areas of saline soil, are revegetation of part of the cleared land with alternative species, pumping to lower the watertable in selected areas, and construction of ditch drains for control of surface water and shallow groundwater. All options are constrained by the economics of dryland farming, and pumping or drainage is further constrained by possible environmental impacts of disposal of saline water.

Application of soil physics/hydrology to salinity in Australia has contributed to understanding, but generally it has proven to be inadequate to aid the development of effective management strategies. A classic approach to soil water movement at the primary catchment scale (areas of order 10⁶ m² or more) will always be limited by errors of measurement at each site within the catchment, those arising from the method of estimation of soil characteristics between measurement sites, and those arising from the method of integration to predict whole-catchment behaviour. The cost and effort of such an approach, and the errors that must eventuate, should be compared with the costs and errors of alternatives for characterization at a whole catchment scale.

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1. Introduction

It is common practice in Australia to measure soil salinity as the chloride concentration (expressed as sodium chloride) in a 1:5 (dry soil to water) extract. Then surface soils (0–0.2 m depth) are said to be

saline if salinity exceeds 0.1% in loams and coarser soils, or 0.2% in clay loams and clays, and sub-soils are said to be saline when salinity exceeds 0.3% (Northcote and Skene, 1972). These criteria approximate an electrical conductivity (EC) of the saturation extract of about 4 dS m⁻¹, which is the criterion for saline soil advocated by the US Salinity Laboratory and used in many countries.

On the basis of these criteria it has been estimated that there are around 386 300 km² of saline soil in

* Corresponding author. Fax: +61-8-9388-1022.

E-mail addresses: ajp@iinet.net.au (A.J. Peck), tom.hatton@per.clw.csiro.au (T. Hatton).

¹ Fax: +61-8-9333-6211.

Australia, or around 5.3% of the area of the continent (Northcote and Skene, 1972).

Secondary salinity is recognized in both irrigated and dryland agricultural areas of Australia. These are areas where surface soil salinity has increased from a non-saline to a saline level as a consequence of irrigation or other agricultural practices. Secondary salinity is common in dryland agricultural areas across southern Australia, and is reported in parts of the US, Canada and Thailand (Salama et al., 1999a,b). In Western Australia (WA) alone the area of farmland affected by secondary salinity is estimated to be 18 000 km², and it is considered that a further 70 000 km² will become saline by the year 2050 (National Land and Water Resources Audit, 2001). This is regarded as the greatest environmental problem in WA.

In broad terms, secondary salinity of non-irrigated farmland in Australia is a consequence of a change of vegetation (generally clearing of native woodland for annual crops and pastures) or land management that has increased groundwater recharge causing a redistribution of soluble salts in the soils. The solutes have accumulated from small inputs in rainfall over a long period of time. In some areas of Australia, and where this problem has been reported in other countries, the salts have been deposited in marine sediments.

Peck and Hurle (1973) used stream gauging and rainfall records, and measurements of the salinity of rainfall to estimate the chloride balance of catchment areas in southwest Australia that remained under natural forest vegetation or had been partly cleared and developed for dryland agriculture. They showed that whereas there was close to a balance between input and output of chloride in the uncleared catchments, the ratio of output to input in partly farmed areas ranged from 3.1 to 21. The increased salt load in streams draining areas of secondary salinity indicates a serious water resource problem. Peck (1973a) collated data to show that at one point on the Blackwood River, with the largest discharge of any river for nearly 2000 km, salinity of streamflow had increased by a factor around 3 over a 50-year period. This environmental disaster has received little attention because only minor use was made of the river water, which is now unsuitable for irrigation or domestic supply. Fig. 1

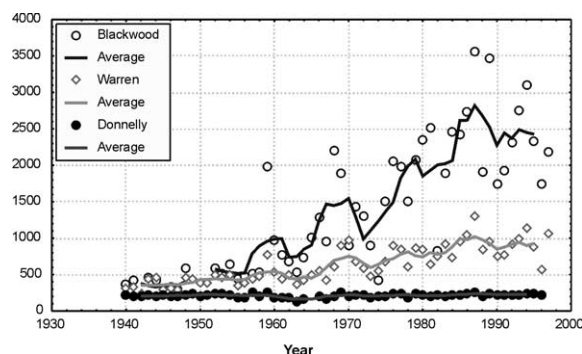


Fig. 1. Trend of salinity for major rivers in southwest Australia. The Blackwood River catchment was extensively cleared in the twentieth century. The upper catchment of the Warren River was also cleared, although to a lesser extent. The Donnelly River catchment is largely intact.

shows the trend of salinity for three major rivers in the region.

Secondary salinity of dryland agricultural areas in WA has been investigated for more than 75 years (Wood, 1924), and the investigations continue. In broad terms, the objectives of this research are to:

- Identify areas at risk of becoming saline. This can be defined as those areas in which there is a probability exceeding a chosen level (e.g. 75%) that the surface soil will become saline within a specified period (e.g. 10 years).
- Develop understanding of the processes that contribute to secondary salinity in sufficient detail to allow reliable selection of options for preventing development, and for reclaiming affected land.
- Evaluate the impact on soil and water salinity of land management procedures that are considered to be potentially viable for control and/or amelioration of secondary salinity at the catchment scale.

This paper presents a review of applications of soil physics/hydrology to the understanding of processes that contribute to secondary salinity in WA. A conceptual model of solute leaching from the complex terrain at the catchment scale is presented, and some major problems that remain to be overcome to meet the above objectives of salinity research are identified.

2. Catchment investigations

2.1. Overview

Many of the investigations of secondary salinity in southwest Australia reviewed in this paper were undertaken in five experimental catchments with areas ranging from 100 to 350 ha that were selected as representative of large areas in the high and intermediate rainfall zones of the region, and in groups in which the physical environment, climate and vegetation are very similar. [Bettenay et al. \(1980\)](#) describe the geology, geomorphology, mineralogy, soils and vegetation of these catchments in detail. They are in two groups of originally forested (native eucalypt) land in areas of Mediterranean climate receiving rainfall of around 1200 and 800 mm a⁻¹. Granitic basement rocks and basic dykes have been deeply weathered (average around 30 m) and lateritised with varying degrees of subsequent erosion and deposition of younger sediments in valleys. The catchments were instrumented in 1973, and in 1977 one whole catchment and large areas of two others were cleared and developed for conventional dryland agriculture.

On the basis of soil and geomorphology mapping, [Bettenay et al. \(1980\)](#) defined four relatively uniform areas of soil, vegetation and landscape position within each catchment. These were defined as 'hydrological provinces' that were used to select sites for instrumentation. At five sites within each catchment, the strata were cored to basement rock or the limit of available equipment for description of the material and determination of bulk density, water content and ionic composition of 1:5 extracts from the samples. Monthly rainfall (storage gauges), soil moisture (neutron access tubes to 6 m depth), groundwater level and EC were recorded at each site. Streamflow and EC were recorded continuously at a gauging site defining each catchment area, and periodically samples of stream and groundwater were taken for chemical analyses. Additional bores were drilled in selected areas to better define groundwater conditions in permanent and ephemeral aquifers. Intensive monitoring continued from 1973 to 1983. Subsequently only flow and EC of surface water discharge from the catchments has been monitored continuously, with the bores monitored at intervals of up to 6 months.

2.2. Soil profile descriptions

[Bettenay et al. \(1980\)](#) describe superficial deposits in the experimental catchments as sands, loams and sesquioxidic gravels from a few centimetres to several metres in depth. In the generalised profile, these soils overlie duricrust, a mottled zone, pallid zone, weathering zone and basement rock although these strata are not always readily distinguished. Commonly they grade into or alternate with one another, particularly mottled zone—pallid zone and pallid zone—weathering zone alternations where the original geology is variable.

The basic rocks weather to produce a regolith with high clay (primarily kaolin) content that may form groundwater barriers. Both granitic and basic rocks weather to form pallid zones with low bulk density (high porosity) where original rock texture is preserved.

2.3. Infiltration characteristics and soil water

[Sharma et al. \(1987a\)](#) measured sorptivity (S) and saturated hydraulic conductivity at saturation (K_s) on regular grids in two of the experimental catchments. The soils have high infiltration capacities under forest conditions, but following clearing and establishment of pasture, K_s decreased by a factor 10, and S by a factor 3. In both cleared and forested catchments, there is little probability of direct runoff from common rainfall events, so most runoff is considered to be generated from areas of soil that are saturated by subsurface flow. Spatial analysis of the infiltration properties did not correspond with the hydrologic provinces defined from mapping of soils and geomorphology.

2.4. Vadose zone moisture regimes

[Sharma et al. \(1987b\)](#) determined soil water regimes under forest and pasture using neutron access tubes to depths of 6 m. The access tubes were in groups of 3 within an area of (nominally) 1000 m² at each of 5 sites in each catchment. Considerable spatial variability of soil water regimes was encountered at each site, between sites within a catchment, and between replicate catchments that received very similar rainfall. The dominant variability was between sites within a catchment. In this environment, rainfall in the winter

months (May–August) results in significant increases of soil water content throughout the measurement depth. Within two years of the change from forest to pasture, there was a statistically significant increase of soil water storage in the profiles whilst due to lower rainfall there was a small decrease in neighbouring uncleared land. It was concluded that increased water storage was a result of reduced evapotranspiration from annual pasture that extracted water from the upper 1 m of the profile relative to the native eucalypt forest that extracted water to depths of at least 6 m.

Turner et al. (1987) determined tritium in vadose zone water. They concluded that there was recent recharge to depths of 7–8 m in a lower rainfall, forested catchment, and throughout the vadose zone (15 m) at a site in a higher rainfall catchment.

2.5. Preferred pathways for water movement

Johnston et al. (1983) irrigated a 10 m² plot with a Rhodamine WT[®] (RWT) tagged NaCl solution at a rate equal to natural rainfall during a period of several months. The plot was on a gentle slope within a naturally forested area. Water samples were taken from monitor bores constructed to sample either a seasonal perched aquifer or the deeper permanent aquifer at this site. Nine months later, the site was excavated to a depth of 8 m to observe the distribution of RWT and Cl on successive vertical faces. The plot overlays an interface between weathered granite and weathered dolerite. It was found that the tracer solution percolated vertically and uniformly through the relatively permeable sandy-gravel surface soil. However, in the underlying pallid zone, RWT staining to depths of 5–5.5 m below ground level showed that the tracer solution followed well-defined vertical and approximately cylindrical channels containing coarser textured material within the weathered granite. Most stained pathways were root channels containing tree roots. Tracer solution also moved around 0.2 m laterally from primary channels to the soil matrix and other soil structures. This was attributed to a combination of mass-flow and diffusion. No vertical flow of the tracer solution was found in the weathered dolerite, which was attributed to a greater abundance of clay and less root penetration in that material.

In a later experiment, Johnston (1987b) measured the vertical distribution of chloride in the vadose zone

to 16 m depth at 13 sites, and groundwater responses in 10 monitor bores and four multilevel piezometers for a period of 2 years within a 700 m² forested plot on one of the higher rainfall experimental catchments. Over most of the plot the vertical flux density of water below the 5 m depth was 2.2–7.2 mm a⁻¹, but within a small portion of the plot the flux density was 50–100 mm a⁻¹ throughout the vadose zone. The latter flux more closely matched the estimated rate of recharge to groundwater. The area of preferred flow was apparently due to a discontinuity within the regolith, resulting in localised recharge to the deep (16 m) watertable within 12–14 h of intense rainfall. A local groundwater mound formed beneath the area of preferred flow, and dissipated over a period of 2–4 days.

2.6. Distribution of soluble salts in the vadose zone

Salt storage in the deeply weathered regolith has been determined from soil cores at many locations in the southwest of WA. Johnston (1987a) refers to these and other studies and presents chloride storage data for the experimental catchments. Amounts ranged from 1.6 to 9.0 kg m⁻² (average 5.0 kg m⁻²) in the higher rainfall catchments and 0.9–38.9 kg m⁻² (average 23.2 kg m⁻²) in the lower rainfall area. Note that the minimum mass of salt (chloride expressed as NaCl) in a soil that is saline to a depth of 1 m is 4.5 kg m⁻² (this follows from the above definition), or 1.8 kg of Cl per square metre. Therefore at most sites that have been studied, vertical redistribution of salt within the profile could result in soil salinity.

Solute distribution in these and other profiles has been classified as ‘bulge’ or ‘monotonically increasing’ with the bulge type being the more common. Fig. 2 presents an example. Assuming one-dimensional (1D) steady flow, chloride ion concentrations in the profiles and in rainfall have been used, together with the moisture-content-dependent diffusion coefficient of chloride in soil to estimate profiles of vertical water flux density. It was concluded that coarse-textured surface soils were well leached by rainfall, but in the underlying more clayey part of the vadose zone the flux of water was very small due to uptake by vegetation. In the lower rainfall region the water flux was typically around 0.4 mm a⁻¹, and in the higher rainfall region it

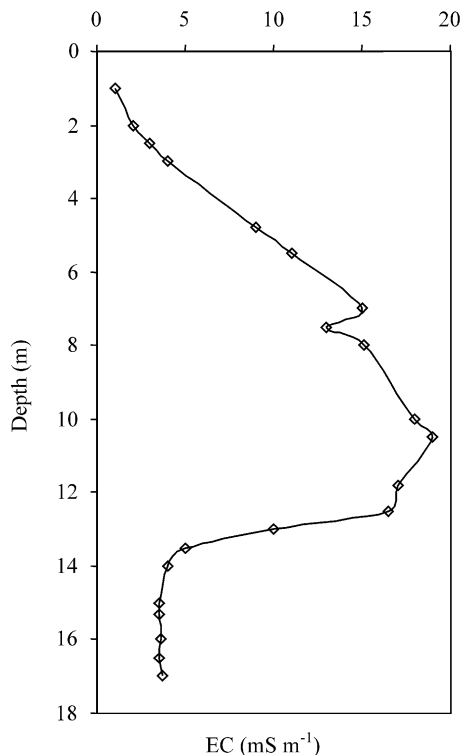


Fig. 2. Example of a typical chloride profile in deeply weathered regolith from southwest Australia (data from Salama et al. (1994)). The bulge in the chloride profile is in the vadose zone (prior to clearing), at a position roughly coincident with the bottom of the rooting zone of native vegetation.

was $2\text{--}5\text{ mm a}^{-1}$. Rates of recharge to the underlying aquifer were estimated to be as much as 100 times greater than the flux in the soil matrix, which was attributed to preferred water flow in structural and textural heterogeneities within the regolith. Preferred pathways were considered to be distributed throughout the landscape, but to constitute only a small part of it.

Chloride ion distribution within the vadose zone was also used by Peck et al. (1981) to estimate the vertical flux density of water at forested sites in WA. The role of diffusion in solute transport was found to be important at three of four sites, and it may dominate convection in a section of one profile. It was concluded that water uptake by roots was restricted to depths of less than 6 m at three of the sites. The shape of the Cl profiles below the depth of maximum concentration could result from a small upward flux of water within the matrix of the deeper regolith whilst

for a short period following significant rainfall events water moved down through preferred pathways.

Environmental isotopes in vadose zone soil moisture were analysed by Turner et al. (1987) who concluded that evaporation from the soil surface plays only a minor role in concentration of solutes in the soil profile. The dominant process in solute accumulation within the vadose zone was ion exclusion during water uptake by plant roots.

2.7. Groundwater recharge

Peck and Williamson (1987) reported results of monitoring groundwater levels in the deeper permanent aquifers (during the winter rainy season, a perched aquifer forms in the more permeable surface soils immediately above the pallid zone clays) within the experimental catchments. In the higher rainfall area, the potentiometric surface increased by 1–4 m during the winter wet season without significant lag, indicating groundwater recharge. Other data (Johnston, 1987a) indicate little flow through the matrix of the vadose zone at these sites. The seasonal response of groundwater level in the lower rainfall catchments was rarely more than 1 m, and at some sites there was no apparent recharge. It was concluded that after a sufficient excess of rainfall over evaporation, there was widespread recharge to the deeper permanent aquifer system.

Following extensive clearing and the establishment of conventional dryland pasture within selected catchments, groundwater levels increased at rates that averaged 2.6 ma^{-1} in both higher and lower rainfall areas. This was interpreted as a result of increased recharge at a rate of $65\text{--}100\text{ mm a}^{-1}$, or around 6–12% of rainfall.

2.8. Aquifer properties

Peck and Williamson (1987) report results of slug test measurements of hydraulic conductivity (K) within the experimental catchments, and Williamson (1986) has collated other results from the region. It has been argued that these results are biased by the method of measurement (George, 1992), and should be slightly larger. However, this does not affect the conclusion that within each catchment or area of investigation, there is variation of K by around a factor

100 with little difference between the geometric mean ($1.4\text{--}7.6\text{ mmd}^{-1}$) in each area studied.

Slug test results indicate the properties of at best a small volume surrounding the borehole. Martin (1984) conducted a conventional pumping test in one of the experimental catchments from which he concluded that the hydraulic conductivity of the weathering zone was 0.27 md^{-1} and the storativity of this aquifer was 0.005. Clarke et al. (1998, 2000) reassessed typical values for K_s in this region and found systematic variations due to geology. These variations are influenced by the presence of dolerite dykes, regional faults, and weathering.

2.9. Catchment-scale water and salt balance and response to clearing

Increases of the yield of water and changes of stream water quality following forest clearing have been documented from locations around the world, but perhaps nowhere as clearly and dramatically as in southwest Australia (Fig. 1). Williamson et al. (1987) report the water and solute balance of the experimental catchments, and the early responses to total or partial clearing for agriculture. Earlier Peck and Hurle (1973) showed that there was a large net export of chloride from river basins in WA that include substantial areas of farmland. There is considerable interest in the dynamics of salt leaching from catchments following agricultural development, since the choice of management strategy, including the 'do nothing' option, may be very different if the return to a new salt balance takes 10 or 100 or 1000 years.

Peck and Hurle (1973) recognised that leaching from a catchment was a multidimensional miscible displacement process for which solutions were available only in simplified situations (Mulqueen and Kirkham, 1972; Peck, 1973b). They estimated the discharge of groundwater using a simple mixing model. Then the characteristic time of catchment leaching is the ratio of the volume of groundwater stored in the catchment to the rate of groundwater discharge. Results for seven farmed catchments in WA ranged from 30 to 400 years.

Fig. 3 presents the annual salt load in streamflow from an experimental catchment in the higher rainfall area. The data were computed from continuous

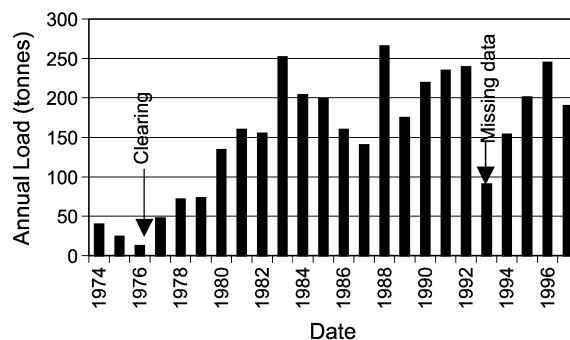


Fig. 3. Flow-weighted mean annual salinity and annual salt loads from a cleared catchment (Wights) in the higher rainfall zone (1200 mm a^{-1}). The catchment was cleared early in 1977 to assess changes in water and salt yield.

records of flow and EC. Macpherson and Peck (1987) describe analysis of the earlier discharge of Cl from this and a neighbouring catchment.

Fig. 3 shows a very large increase of salt discharge from the catchment in the seven years following clearing. Subsequently variation of rainfall from year to year, and missing data in 1993 mask any increasing or decreasing trend. Using the leaching model of Mulqueen and Kirkham (1972), the characteristic time of leaching from this catchment was estimated to be about 10 years. However, more than 20 years post-clearing there is no indication of significant decay of stream salt load. Raats (1978, 1981) addressed the multidimensional leaching problem and showed that in the case of steady flow through a uniform 2D (section) of aquifer, the exponential decay process described by Mulqueen and Kirkham (1972) and Peck (1973b) is limited to cases where the ratio of aquifer length to thickness is greater than about 5. Whilst the Raats model provides valuable insights to the process of solute transport within a catchment, its applicability is limited by assumptions of steady flow and uniform aquifer properties.

Possible reasons for failure of the exponential leaching model in this catchment are non-stationarity of the groundwater system, aquifer layering (i.e. variations of hydraulic conductivity with depth in the strata) and solute exchange between the matrix and preferred pathways for water movement in the mottled, pallid and weathering zones.

3. A conceptual model for saline catchments in southwestern Australia

The process of understanding the structure, function and response of the southwest Australian catchments prone to salinisation must be captured within an appropriate conceptual model before we can move toward analytical or numerical simulations or predictions of their behaviour. The hydrological and hydrogeological conceptualisation of these catchments has a reasonably long history (Wood, 1924; Bettenay and Mulcahy, 1972; Peck et al., 1980; George, 1992; George and Conacher, 1993a,b; Salama, 1997; Clarke et al., 1998, 2000). Salama et al. (1993a,b,c, 1994) gave the most detailed description of the controls and processes related to salt mobilisation at the catchment scale; the implied complexity of the spatial and temporal response of salt yield to clearing is great (Fig. 4). Peck and Williamson (1987) concluded that the large temporal and spatial variability in groundwater salinity could not be interpreted within the context of a solute transport equation.

Modelling the behaviour of salt in a catchment in disequilibrium depends on the complexity and character of the conceptual model of salt storage,

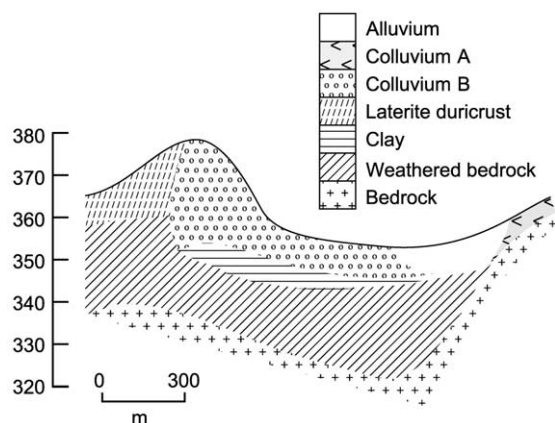


Fig. 4. A conceptual model of the hydrogeology of extensively cleared catchments on deeply weathered basement rocks in southwest Australia, from Salama et al. (1993a). After clearing a permanent watertable within the weathered bedrock rises until this aquifer is semi-confined by the clay. This aquifer discharges most of the salt reaching the soil surface and streams. A transient, perched watertable develops within colluvium above the clay, and delivers seasonal throughflow to the stream.

mobilisation, and delivery to the stream. While one can imagine any level of complexity in the representation of a landscape yielding salt, and subsequently develop the computational means to forward model salt yield, the challenge is to abstract the yield processes to a degree that is roughly commensurate with available data and knowledge (Hatton et al., 1994). Thus at one extreme, models using quasi-realistic physical modelling (e.g. WEC-C: Croton and Bari, 2001) explicitly track salt and water fluxes through multiple layers with spatially distributed vegetation, regolith and climatic variables. Estimation of the parameters of such models is an extreme inverse modelling problem, with serious issues including non-unique specification and strong cross-correlation of estimates (Allison and Peck, 1985; Macpherson and Peck, 1987; Wheater et al., 1993). The adequacy of the conceptualisation and parameterisation of such models is moot in the sense that effectively there is no catchment in Australia where there is enough data to constrain or test such models toward the point of their validation. The fact that such complex models cannot be validated in practice (*sensu* Oreskes et al., 1994) obviates their use in real-world prediction. Complex models of salt redistribution and yield can be useful for testing our ideas about the real world based on an idealised (hypothetical) formulation (*sensu* Beven, 1993), but in this sense their value tends to lie in the discovery of inadequacies in our conceptualisation or mathematics rather than any success in mimicking observations. Further, the limited physical and chemical detail available across the Australian landscape at risk from salinisation is such that simpler models of salt yield are demanded and appropriate.

The balance between data availability, process understanding and model complexity is exemplified in Hookey (1987), who modelled the development of groundwater levels and salinisation in a small first-order catchment in Western Australia. In this case, a numerical representation of the nonsteady, 2D groundwater flow in an unconfined, nonhomogeneous and isotropic aquifer was parameterised for a partially cleared catchment based on limited test-pumping data, assumed recharge values, and a very simplified conceptual model of flow. Despite this relatively simple model realisation, calibration against stream-flow and bore trends required substantial changes in

parameter values. This analysis defined the time to hydrological equilibrium between recharge and discharge; to estimate the persistence of saline discharge, a simplified exponential decay model was invoked, based on the ratio of groundwater discharge to the regolith water storage term.

4. Prediction of salt discharge from a catchment

Peck (1976) inverted a method used to estimate additional groundwater recharge beneath cleared land from streamflow records to allow prediction of stream salinity following agricultural development. This method was subsequently modified to predict the dynamic response of stream salinity to progressive clearing and possibly revegetation within several sub-catchments of a river basin (Peck et al., 1977). The latter model includes a generalised representation of the groundwater system to provide a time lag between clearing part of a catchment and additional discharge of groundwater to a gaining stream.

Schofield (1986) and Schofield et al. (1988) report other models that have been developed to estimate effects of clearing on stream salinity in WA. The first uses empirical relationships between clearing, streamflow and salt load within sub-catchments to investigate various clearing scenarios. The second extends the simple mixing model applied by Peck (1976) to include a second aquifer with lower salinity.

In their analysis of salt discharge from Wights catchment (Fig. 3), Macpherson and Peck (1987) considered three mathematical models of the leaching process. The first was a simple piston flow model, the second a first-order kinetic model that took no account of catchment geometry, and the third took some account of the initial spatial distribution of salt within the catchment. These models take no explicit account of physical processes that are believed to occur during catchment leaching. Model parameters were estimated by fitting to the measured data. Applying the first model, the fitting parameter did not converge to a stable value, but it was concluded that the expected time to chloride equilibrium in this catchment was 64 ± 50 years. Whilst one parameter of the second model could be determined with exceptional precision, the second was definitely non-stationary. It appeared that leaching became more efficient with time. Using

median values of the parameters, excess salt movement was estimated to reach 37% of its final value in 45 years and 87% after about 90 years. The third model also failed to give stationary estimates of the parameters.

5. Prediction of areas at risk of salinity

In the broadest view, most of the southern Australian landscape in the annual rainfall range of 250–800 mm with deeply weathered regolith has the potential for salinisation following clearing. However, within that zone there is significant variation in the degree of salinisation, and its persistence, among hydrogeomorphic units. This can range from mild land salinisation with marginal impacts on crop production to the development of extensive salt flats essentially devoid of vegetation. Similar variation extends to the degree of stream salinisation (National Land and Water Resources Audit, 2001). Understanding the source of this variation is fundamental to predicting the local risk.

The standard approach to prediction of salinity risk in Australia is based on inferring the ultimate extent of shallow watertables and their rate of development; assuming that a shallow watertable (typically 1 or 2 m below ground level) brings with it the risk of salinisation (Nulsen, 1981). Using this criterion together with forecasts of equilibrium watertable positions based on historic trends, the National Land and Water Resources Audit (2001) concluded that the area with a high risk of land salinisation might increase to 17 million ha in the future.

In reality, it is recognised that a shallow watertable is neither a necessary nor sufficient condition for land and stream salinisation. Controls on the near-surface salt balance are reasonably well understood in theoretical terms, but difficult to predict in practice. The vagaries of multiple interacting aquifers, aquifer confinement, shallow throughflow, surface wash off, and seasonally fluctuating groundwater level can all combine to mitigate or intensify the surface salt balance. It is highly unlikely that Australia will ever have 17 million ha of severely salt-affected land.

Predicting groundwater levels in catchments in disequilibrium is challenging enough without the added complexities of the salt balance. Hatton et al. (1994) review the diverse modelling approaches used

for this purpose. In Australia, the most common approach is to apply a standard two or 3D groundwater model in which aquifer properties and recharge are specified in time and space, and watertables (or heads) and discharge are predicted. Examples include Hookey (1987) and Salama et al. (1993c). Hatton (1998) reviewed catchment scale groundwater recharge models. The key feature here is the specification of a long-term average recharge rate associated with a particular land use. Other modelling approaches endeavour to incorporate soil-plant-atmosphere transfer algorithms (and even plant growth algorithms) to model recharge as a function of climate and land use. Examples are TOPOG (Dawes et al., 1997; Zhang et al., 1999a) and WEC-C (Croton and Bari, 2001).

Ultimately, real-world salinity prediction must be based on a reasonable conceptual model of the system in question and the relevant available data for that system. Using this approach, Peck and Hurlé (1973) and Salama et al. (1993a) predicted the development or persistence of saline discharge on the basis of known storages of groundwater and salt, salt inputs and discharge rates. The key challenges that arise from these rational approximations are the estimation of the degree of mixing of groundwaters of varying salinities, and the estimation of the bulk salt store at the catchment scale. Advances in down hole (Salama et al., 1994) and airborne (George and Bennett, 2000) electromagnetic induction techniques offer some hope that this latter key below ground parameter can be estimated and mapped with increased accuracy.

Another approach to model improvement is to develop the theoretical relationships among parameters that are presently treated as independent in our models. On the hydraulic side of the problem, Salama et al. (1997) and Salama et al. (1999a) argued that the evolution of land systems in Australia lead to strong covariation of topography, groundwater depths and gradients, and aquifer properties, based on earlier ideas of Strahler (1956) and Toth (1963, 1978). These authors introduced and applied hydrogeomorphic principles to the problem of parameter specification in salinising groundwater systems. This approach offers a way forward from the pernicious problem of parameter specification in aquifer models.

On the water balance side of the modelling challenge, a similar approach to parameter minimisation

through the introduction of theoretical relationships among modelling terms was developed by Eagleson (1982). In this work, ecohydrological theory relating soil moisture, vegetation biomass, and drainage fluxes was used to eliminate below ground physical parameters notoriously difficult to measure at appropriate scales. Hatton et al. (1997) reviewed this approach and contrast it with standard hydrological modelling approaches based on unconstrained relationships among parameter values.

These geomorphological and ecohydrological approaches to the identification of emergent system properties indicate one way forward for the challenge of predicting the salinity behaviour of catchments in disequilibrium. The theoretical constraints proposed by Salama and Eagleson apply, but it is also likely that additional feedback between vegetation, landscape and salt will lead to yet more emergent behaviour that can provide further constraints on model formulation and parameter estimation.

6. Problems relating to salinity management

In Australia, dryland salinity management tends to resolve into three different approaches: recharge control, discharge enhancement, and living with saline land and water. By far the bulk of effort has been directed at the former, with immense public and private investment in tree planting and the search for new low recharge farming systems (see the collection of papers in Holmes and Talsma (1981)). However, Hatton and Salama (1999), George et al. (2001) and Clarke et al. (2002) indicate that the likelihood of reversing or even containing salinity and recovering rivers through recharge control is low over much of the area currently or potentially affected. Our limited ability to reverse the hydrological and hydrogeochemical processes put in train with vegetation clearing raises profound technical, sociological and political issues.

The control of recharge is complicated by the apparent episodicity of recharge events across much of Australia's salt affected region (Zhang et al., 1999b), both naturally and after clearing. Long-term average recharge rates, as estimated by time-integrating methods (Walker et al., 1991), belie the actual amounts and frequency with which recharge occurs. The implications of this episodicity to recharge control

are immense; low-recharge control farming systems have to aspire to preventing these large, occasional events, not a regular, smaller average target.

Engineering alternatives that enhance drainage and dispose of the saline effluent are a clear alternative and in fact have long been a feature of Australia's response to salinity hazard mitigation with respect to high value assets at risk. This engineering can take a variety of forms including surface water control, groundwater pumping, groundwater drains, and disposal basins. The specification and design of these works, however, is beset by many of the same problems that plague predictive modelling. From place to place, it is difficult to forecast the effectiveness of these remedial approaches to land and water salinity remediation. Given their known cost, the uncertainty of benefits clouds their potential role in salinity management. The national recognition that in many cases only engineering intervention can hope to mitigate salinity impacts (National Land and Water Resources Audit, 2001) adds significant impetus for the requisite research and development to minimise this uncertainty.

Australia is also coming to realise that regardless of what we as a nation do, we will have significantly more salinised water and land into the future. This realisation has turned attention to the development of new uses for saline resources, including saline farming systems. Optimism for the potential for such systems, however, is mitigated by observations that vegetation growing in the presence of high saline watertables may not be sustainable due to salt accumulation in the root zone (Thorburn et al., 1995; Slavich et al., 1999). Elucidation of those processes that maintain a favourable root zone salt balance in saline discharge areas is crucial for minimising the risk of failure in these dryland saline-farming enterprises. Hatton (1999) proposed looking for this understanding by investigating the factors that make natural saline ecosystems sustainable in such environments.

7. Conclusions

Secondary salinity following clearing of native vegetation for dryland agriculture is a serious environmental and economic problem affecting soil and water in Australia. In Western Australia,

increased recharge to an aquifer in partly weathered basement rock results in development of areas where groundwater is lost by evapotranspiration or discharge to a stream. A fundamental problem for salinity risk assessment and management is the broad-acre estimation of recharge rate averaged over an area of at least 1 ha where vertical movement is mostly through remnant root paths and rock structures in strata above the aquifer. The quantity of soluble salts that has accumulated in the vadose zone of this complex medium by natural processes is sufficient to result in soil salinity when redistributed by diffusion to more permeable structures, and upward convection of soil water to discharge areas.

Substantial contributions have been made to understanding water and solute storage and transport, but the complexity of this landscape currently defies the compass of predictive soil physics and hydraulics. Developments in prediction have been based largely on empirical extrapolations or the mathematics of simplified conceptualisations. Whilst such simplified analyses are perhaps robust, they are at best proximate and they do not lead to a deeper understanding of process.

The challenge to marry what we know about fine-scale processes with landscape complexity and response remains. It may be that the identification of emergent properties at field and catchment scales will lend themselves to process interpretations based on conventional soil physics and hydraulics, but we have a considerable way to go to establish this connection.

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